

Topics Covered:

- Data organization & Data presentation

1. Fifty people were asked to identify their favorite sport. The responses were F professional football, C college football, B baseball, and A auto racing. Construct a categorical frequency distribution for the data. Which class has the most data values, and which class has the fewest data values?

F	C	F	A	B
C	F	F	B	C
A	B	C	C	F
F	A	F	C	A
C	C	F	F	C
F	F	B	C	F
B	A	F	B	F
F	C	B	A	C
B	F	C	F	A
B	B	F	C	F

Answer:

Class	Tally	Frequency	Percent
F	 	19	38
C	 	14	28
B	 	10	20
A	 	7	14
		Total 50	100

The class with the most data values is professional football.

The class with the least number of data values is auto racing.

2. Forty plasma TVs were tested, and the number of watts per hour that they used while running for 4 hours per day were recorded. Construct a frequency distribution for the data. Use 6 classes.

480	503	514	472	465
473	475	500	502	506
492	477	483	475	481
512	473	488	485	503
508	484	486	499	512
485	468	507	501	470
497	479	492	468	492
466	472	483	511	481

Answer:

Limits	Boundaries	f	cf
465–473	464.5–473.5	9	Less than 464.5 0
474–482	473.5–482.5	7	Less than 473.5 9
483–491	482.5–491.5	7	Less than 482.5 16
492–500	491.5–500.5	6	Less than 491.5 23
501–509	500.5–509.5	7	Less than 500.5 29
510–518	509.5–518.5	4	Less than 509.5 36
		Total 40	Less than 518.5 40



3. A sample of 40 large cities was selected, and the average of the wind speeds was computed for each city over one year. Construct a frequency distribution, using 7 classes.

12.2	9.1	11.2	9.0
10.5	8.2	8.9	12.2
9.5	10.2	7.1	11.0
6.2	7.9	8.7	8.4
8.9	8.8	7.1	10.1
8.7	10.5	10.2	10.7
7.9	8.3	8.7	8.7
10.4	7.7	12.3	10.7
7.7	7.8	11.8	10.5
9.6	9.6	8.6	10.3

Answer:

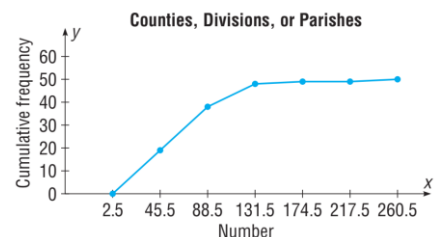
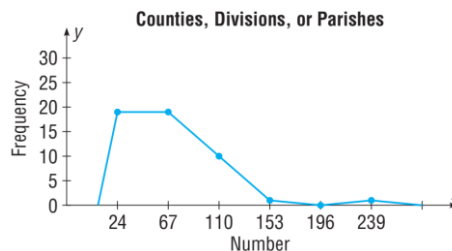
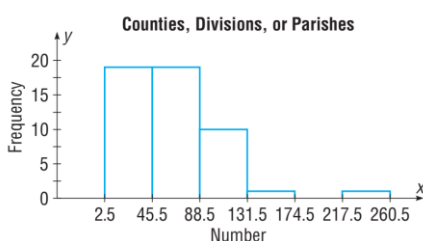
Limits	Boundaries	f		cf
6.2–7.0	6.15–7.05	1	Less than 6.15	0
7.1–7.9	7.05–7.95	7	Less than 7.05	1
8.0–8.8	7.95–8.85	9	Less than 7.95	8
8.9–9.7	8.85–9.75	7	Less than 8.85	17
9.8–10.6	9.75–10.65	8	Less than 9.75	24
10.7–11.5	10.65–11.55	4	Less than 10.65	32
11.6–12.4	11.55–12.45	4	Less than 11.55	36
		Total	Less than 12.45	40
		40		

4. The number of counties, divisions, or parishes for each of the 50 states is given below. Use the data to construct a grouped frequency distribution with 6 classes, a histogram, a frequency polygon, and an ogive. Analyze the distribution.

67	27	15	75	58	64	8	67	159	5
102	44	92	99	105	120	64	16	23	14
83	87	82	114	56	93	16	10	21	33
62	100	53	88	77	36	67	5	46	66
95	260	29	14	95	39	55	72	23	3

Answer:

Limits	Boundaries	f	Midpoints		cf
3–45	2.5–45.5	19	24	Less than 2.5	0
46–88	45.5–88.5	19	67	Less than 45.5	19
89–131	88.5–131.5	10	110	Less than 88.5	38
132–174	131.5–174.5	1	153	Less than 131.5	48
175–217	174.5–217.5	0	196	Less than 174.5	49
218–260	217.5–260.5	1	239	Less than 217.5	49
		Total	50	Less than 260.5	50



5. The frequency distribution shows the total number of points scored in the Super Bowl games from 1967 to 2012. Construct a histogram, frequency polygon, and ogive for the distribution. If you were to predict the total number of points for the next Super Bowl game, what range of values would you choose? In how many games were the total points greater than 56?

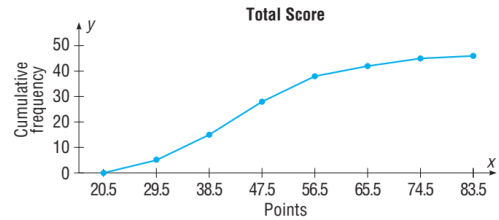
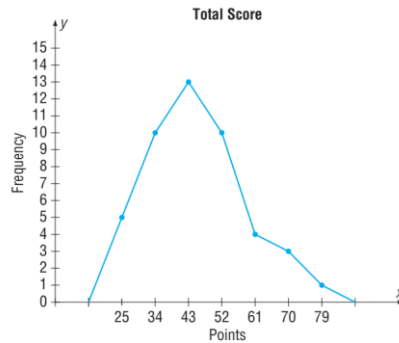
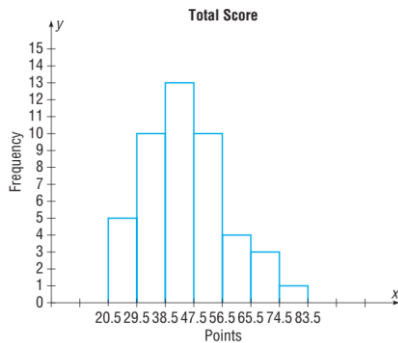
Class limits	Frequency
21–29	5
30–38	10
39–47	13
48–56	10
57–65	4
66–74	3
75–83	1
Total	46

Answer:

Limits	Boundaries	f	Midpoints		cf
21–29	20.5–29.5	5	25	Less than 20.5	0
30–38	29.5–38.5	10	34	Less than 29.5	5
39–47	38.5–47.5	13	43	Less than 38.5	15
48–56	47.5–56.5	10	52	Less than 47.5	28
57–65	56.5–65.5	4	61	Less than 56.5	38
66–74	65.5–74.5	3	70	Less than 65.5	42
75–83	74.5–83.5	1	79	Less than 74.5	45
		1		Less than 83.5	46
	Total	46			

Eight games had more than 47.5 goals scored.

Eight games had more than 56 total points.



6. For 108 randomly selected college applicants, the following frequency distribution for entrance exam scores was obtained. Construct a histogram, frequency polygon, and ogive for the data. Construct a histogram, frequency polygon, and ogive, using relative frequencies.

Class limits	Frequency
90–98	6
99–107	22
108–116	43
117–125	28
126–134	9
Total	108



Answer:

Limits	Boundaries	<i>f</i>	Midpoints
90–98	89.5–98.5	6	94
99–107	98.5–107.5	22	103
108–116	107.5–116.5	43	112
117–125	116.5–125.5	28	121
126–134	125.5–134.5	9	130
		Total	108

	<i>cf</i>
Less than 89.5	0
Less than 98.5	6
Less than 107.5	28
Less than 116.5	71
Less than 125.5	99
Less than 134.5	108

Eighty applicants do not need to enroll in the developmental programs.

